

MIXED HODGE STRUCTURES AND EQUIVARIANCE: FROM THE DOUBLE COMPLEX TO THE WEIGHT-GRADED ACTION

This note follows the column-filtration convention of the spectral-sequence notes [SS]. The logical order is

$$(K, \delta_1, \delta_2) \longmapsto (C^\bullet, D, F) \longmapsto (E_r, d_r) \longmapsto \mathrm{Gr}_F H^*(C^\bullet, D).$$

First-quadrant convergence is an algebraic result. Hodge theory will additionally imply degeneration at E_2 and identify the resulting quotients with weight-graded Hodge structures. We reserve F for the column filtration and its induced filtration on total cohomology, use L for its transport to simplicial cohomology, and write F_{Hdg} for the Hodge filtration.

1. THE FIRST-QUADRANT DOUBLE COMPLEX

Definition 1.1 (Set up and cochains). Let $X_\bullet$ be a simplicial complex scheme such that each X_p is a finite disjoint union of smooth projective varieties. Write $d_i^{(p)} : X_p \rightarrow X_{p-1}$, $0 \leq i \leq p$, for its face maps. Let $f_\bullet : X_\bullet \rightarrow X_\bullet$ be a simplicial endomorphism with components f_p ; thus $d_i^{(p)} f_p = f_{p-1} d_i^{(p)}$. Define

$$K^{p,q} := C_{\mathrm{sing}}^q(X_p^{\mathrm{an}}, \mathbb{Q}) \quad (p, q \geq 0), \quad K^{p,q} := 0 \quad \text{otherwise,}$$

and

$$\begin{aligned} \delta_1^{p,q} &:= \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^* : K^{p,q} \longrightarrow K^{p+1,q}, \\ \delta_2^{p,q} &:= (-1)^p d_{\mathrm{sing}} : K^{p,q} \longrightarrow K^{p,q+1}. \end{aligned}$$

Set $C^n := \mathrm{Tot}^n K = \bigoplus_{p+q=n} K^{p,q}$ and $D := \delta_1 + \delta_2$.

The simplicial identities give $\delta_1^2 = 0$. Pullback commutes with d_{sing} , and the sign in δ_2 changes when δ_1 raises p . Hence

$$\delta_2^2 = 0, \quad \delta_1 \delta_2 + \delta_2 \delta_1 = 0, \quad D^2 = 0.$$

This is a first-quadrant double cochain complex in exactly the convention of the spectral-sequence notes. We use its total complex as the rational cochain model for simplicial cohomology:

$$H_X^n := H^n(X_\bullet, \mathbb{Q}) = H^n(C^\bullet, D) =: H_C^n.$$

This concerns $X_\bullet$, not an additional variety X . An identification with $H^n(X, \mathbb{Q})$ would require an augmentation and a cohomological descent theorem.

Definition 1.2 (Subcomplexes and induced filtration). For each integer p put

$$F^p C^n := \bigoplus_{i \geq p} K^{i, n-i}, \quad D_p := D|_{F^p C^\bullet}.$$

Since $D(F^p C^n) \subseteq F^p C^{n+1}$, the inclusion $\iota_p : (F^p C^\bullet, D_p) \rightarrow (C^\bullet, D)$ is a cochain map. Define

$$H^n(\iota_p) : [a]_{D_p} \longmapsto [a]_D, \quad F^p H_C^n := \mathrm{im} H^n(\iota_p).$$

Let $L^p H_X^n$ be the image of $F^p H_C^n$ under $H_C^n = H_X^n$. Thus L is transported from F , not chosen independently.

For $n \geq 0$, $F^0 C^n = C^n$ and $F^{n+1} C^n = 0$, so

$$H_C^n = F^0 H_C^n \supseteq \cdots \supseteq F^{n+1} H_C^n = 0.$$

A class lies in $F^p H_C^n$ exactly when it has a closed representative in $F^p C^n$. We write $\mathrm{Gr}_F^p H_C^n := F^p H_C^n / F^{p+1} H_C^n$, and similarly for L .

2. PAGES, DIFFERENTIALS, AND CONVERGENCE

Definition 2.1 (The representative-quotient convention). For $n = p + q$ and integer r , put

$$Z_r^{p,q} := F^p C^n \cap D^{-1}(F^{p+r} C^{n+1}), \quad B_r^{p,q} := F^p C^n \cap D(F^{p-r} C^{n-1}).$$

For the pages $r \geq 0$ define

$$E_r^{p,q} := \frac{Z_r^{p,q}}{Z_{r-1}^{p+1,q-1} + B_{r-1}^{p,q}}, \quad [a]_r := a + (Z_{r-1}^{p+1,q-1} + B_{r-1}^{p,q}),$$

$$d_r^{p,q} : E_r^{p,q} \longrightarrow E_r^{p+r,q-r+1}, \quad [a]_r \longmapsto [Da]_r.$$

Negative auxiliary subscripts make this quotient formula valid also at $r = 0$; we do not assert a spectral sequence of negative pages.

Here a is a total cochain, not necessarily an element of the single summand $K^{p,q}$. The denominator is contained in $Z_r^{p,q}$. Moreover $D(Z_r^{p,q}) \subseteq Z_r^{p+r,q-r+1}$, and D takes the source denominator into the target denominator. Thus the displayed d_r is well-defined and $d_r^2 = 0$. The next-page proposition of [SS] gives the canonical isomorphism

$$(1) \quad \Psi_{r+1}^{p,q} : E_{r+1}^{p,q} \xrightarrow{\sim} \ker d_r^{p,q} / \mathrm{im} d_r^{p-r,q+r-1},$$

$$[a]_{r+1} \longmapsto [a]_r + \mathrm{im} d_r^{p-r,q+r-1}.$$

This is a property of the constructed pages, not their definition. In general it gives no map from all of E_r to E_{r+1} .

Proposition 2.2 (First pages). *For the double complex of Section 1, there are canonical identifications*

$$E_0^{p,q} \cong \mathrm{Gr}_F^p C^{p+q} \cong K^{p,q}, \quad d_0 = \delta_2,$$

$$E_1^{p,q} \cong H^q(X_p, \mathbb{Q}), \quad d_1^{p,q} = \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^*.$$

Then

$$E_2^{p,q} \cong \frac{\ker(d_1^{p,q} : H^q(X_p, \mathbb{Q}) \rightarrow H^q(X_{p+1}, \mathbb{Q}))}{\mathrm{im}(d_1^{p-1,q} : H^q(X_{p-1}, \mathbb{Q}) \rightarrow H^q(X_p, \mathbb{Q}))},$$

with zero denominator when $p = 0$.

Proof. We have $Z_0^{p,q} = F^p C^n$, $Z_{-1}^{p+1,q-1} = F^{p+1} C^n$, and $B_{-1}^{p,q} \subseteq F^{p+1} C^n$. This proves the formula for E_0 . Projection to column p identifies d_0 with δ_2 , since δ_1 raises the column. Multiplying d_{sing} by $(-1)^p$ does not change its kernels or images, so its cohomology is $H^q(X_p, \mathbb{Q})$. For a vertical cocycle $a \in K^{p,q}$, $Da = \delta_1 a$; thus d_1 is the displayed alternating pullback. Apply (1) to obtain E_2 . $\square$

Theorem 2.3 (Convergence for the induced filtration). *Take the double complex and column filtration of Section 1. Then the spectral sequence (E_r, d_r) converges to $H^*(C^\bullet, D)$ with its specified induced filtration F : for $n = p + q$ there is a canonical isomorphism*

$$\theta^{p,q} : E_\infty^{p,q} \xrightarrow{\sim} \mathrm{Gr}_F^p H_C^n, \quad [a]_\infty \longmapsto [a]_D + F^{p+1} H_C^n.$$

Transporting along $H_C^n = H_X^n$ gives $E_\infty^{p,q} \cong \mathrm{Gr}_L^p H_X^n$.

Proof. Fix $p, q \geq 0$ and choose $r > \max\{p, q + 1\}$. Then

$$F^{p+r}C^{n+1} = 0, \quad F^{p-r+1}C^{n-1} = C^{n-1}.$$

Put $Z_\infty^{p,q} := F^p C^n \cap \ker D$ and $B_\infty^{p,q} := F^p C^n \cap \operatorname{im} D$. The defining quotient becomes

$$E_r^{p,q} = \frac{Z_\infty^{p,q}}{Z_\infty^{p+1,q-1} + B_\infty^{p,q}}.$$

This holds on every later page, and the natural identifications preserve representatives. The common stable space is $E_\infty^{p,q}$. The map $a \mapsto [a]_D + F^{p+1}H_C^n$ from $Z_\infty^{p,q}$ onto $\operatorname{Gr}_F^p H_C^n$ is surjective by the definition of $F^p H_C^n$. Its value is zero precisely when

$$a = b + Dc, \quad b \in F^{p+1}C^n, \quad Db = 0, \quad c \in C^{n-1}.$$

Since $a, b \in F^p C^n$, its kernel is exactly $Z_\infty^{p+1,q-1} + B_\infty^{p,q}$. Taking the quotient proves the theorem. $\square$

Each $E_1^{p,q}$ is finite-dimensional, and only finitely many bidegrees lie on a fixed total diagonal. Convergence and the finite target filtration therefore also prove that H_X^n is finite-dimensional.

Remark 2.4 (Which spectral sequence in McCleary?). This is the first, column-filtered spectral sequence ${}^I E$ of McCleary's Theorem 2.15, obtained from Theorem 2.6 [McC]. His Definition 2.4 asks for the existence of a suitable target filtration. Here, as in our notes, we specify the induced filtration in advance and prove the comparison for it. Neither projectivity nor a Hodge structure is used in Theorem 2.3.

3. HODGE AND WEIGHT FILTRATIONS: THE GEOMETRIC INPUT

For the Hodge filtration form a separate total complex of smooth forms:

$$\mathcal{B}^m := \bigoplus_{p+a+b=m} \mathcal{A}^{a,b}(X_p), \quad D_{\mathcal{B}} := \delta_1 + (-1)^p(\partial + \bar{\partial}).$$

All three indices are nonnegative; δ_1 is the alternating sum of face pullbacks on forms. The natural de Rham comparison gives

$$c_n : H^n(\mathcal{B}^\bullet, D_{\mathcal{B}}) \xrightarrow{\sim} H_X^n \otimes_{\mathbb{Q}} \mathbb{C} =: H_{X,\mathbb{C}}^n.$$

It respects simplicial pullbacks and the filtration by p . One may construct it through smooth singular cochains. No identification of rational cochain spaces tensored with $\mathbb{C}$ with all complex cochains is required.

Definition 3.1 (The two filtrations of the MHS). Set

$$F_{\text{Hdg}}^\ell \mathcal{B}^m := \bigoplus_{\substack{p+a+b=m \\ a \geq \ell}} \mathcal{A}^{a,b}(X_p).$$

These are subcomplexes: holomorphic pullbacks preserve type, and $\partial + \bar{\partial}$ does not decrease a . Define

$$F_{\text{Hdg}}^\ell H_{X,\mathbb{C}}^n := c_n \left(\operatorname{im} \left(H^n(F_{\text{Hdg}}^\ell \mathcal{B}^\bullet) \rightarrow H^n(\mathcal{B}^\bullet) \right) \right),$$

using the restrictions of $D_{\mathcal{B}}$. On rational cohomology set

$$W_w H_X^n := L^{n-w} H_X^n \quad (w \in \mathbb{Z}).$$

This filtration is increasing, with $W_{-1}H_X^n = 0$ and $W_n H_X^n = H_X^n$. For $n = p + q$, the index conversion is

$$(2) \quad \operatorname{Gr}_q^W H_X^n = \frac{L^p H_X^n}{L^{p+1} H_X^n} = \operatorname{Gr}_L^p H_X^n \cong \operatorname{Gr}_F^p H_C^n.$$

Recall that a rational mixed Hodge structure is a finite-dimensional $\mathbb{Q}$ -space V with finite filtrations W (increasing) and F_{Hdg} (decreasing on $V_{\mathbb{C}}$), such that each $\text{Gr}_w^W V$ is pure of weight w . For a pure weight- w space U , this means

$$U_{\mathbb{C}} = \bigoplus_{a+b=w} U^{a,b}, \quad \overline{U^{a,b}} = U^{b,a}, \quad F_{\text{Hdg}}^{\ell} U_{\mathbb{C}} = \bigoplus_{a \geq \ell} U^{a,w-a}.$$

On the weight quotient, the Hodge filtration is induced by that on V :

$$F_{\text{Hdg}}^{\ell}(\text{Gr}_w^W V)_{\mathbb{C}} = \frac{(F_{\text{Hdg}}^{\ell} V_{\mathbb{C}} \cap W_w V_{\mathbb{C}}) + W_{w-1} V_{\mathbb{C}}}{W_{w-1} V_{\mathbb{C}}}.$$

A morphism is a rational linear map preserving both filtrations.

Theorem 3.2 (Geometric input: Deligne). *Let $X_{\bullet}$ be as in Section 1. Then $(H_X^n, W, F_{\text{Hdg}})$, with the filtrations just defined, is its natural mixed Hodge structure. The simplicial spectral sequence from E_1 onward is a spectral sequence of mixed Hodge structures: its differentials, next-page identifications, and convergence comparisons respect these structures. On $E_1^{p,q}$ the structure is the usual pure weight- q structure on $H^q(X_p, \mathbb{Q})$. These constructions are natural for simplicial morphisms.*

This is the external geometric input from Deligne, applied to smooth projective levels with empty boundary [D74, (8.1.19)–(8.1.20), (8.3.5)]. It is not a consequence of the algebraic convergence theorem. In particular, purity of E_1 alone would not justify regarding all higher differentials as Hodge morphisms; their compatibility is part of this input. The following arguments prove the algebraic consequences we need.

4. PURITY IMPLIES DEGENERATION AT THE SECOND PAGE

Lemma 4.1. *Let $u : V \rightarrow V'$ be a morphism of mixed Hodge structures, where V, V' are pure of weights w, w' . If $w \neq w'$, then $u = 0$.*

Proof. If $w < w'$, then $u(V) = u(W_w V) \subseteq W_w V' = 0$. If $w > w'$, take $x \in V^{a,b}$ with $a + b = w$. The map $u_{\mathbb{C}}$ preserves F_{Hdg} and, since it is rational, its conjugate. Thus

$$u_{\mathbb{C}}(x) \in F_{\text{Hdg}}^a V'_{\mathbb{C}} \cap \overline{F_{\text{Hdg}}^b V'_{\mathbb{C}}} = \bigoplus_{\substack{i+j=w' \\ i \geq a, j \geq b}} V'^{i,j} = 0,$$

since $w' < a + b$. This proves the assertion in both cases. $\square$

Proposition 4.2 (Degeneration). *Under the setup of Section 1, $E_2^{p,q}$ is pure of weight q . Then $d_r^{p,q} = 0$ for all $r \geq 2$ and all p, q , and the natural next-page identifications give*

$$\sigma_{p,q} : E_2^{p,q} \xrightarrow{\sim} E_{\infty}^{p,q}.$$

Proof. The maps d_1 connect pure structures of the same weight q and preserve their Hodge-type decompositions. Their kernels and images are therefore sums of type components, and $E_2^{p,q}$ is pure of weight q . Suppose inductively that $E_r^{p,q}$ is pure of weight q for every p, q . By Theorem 3.2,

$$d_r^{p,q} : E_r^{p,q} \longrightarrow E_r^{p+r, q-r+1}$$

is a Hodge morphism. For $r \geq 2$ the source and target have different weights q and $q - r + 1$, so Lemma 4.1 makes it zero. Equation (1) identifies E_{r+1} with E_r as Hodge structures, completing the induction. Compose the natural identifications up to the stable page in each bidegree to obtain $\sigma_{p,q}$. $\square$

Combining this with Theorem 2.3 and (2) gives

$$E_2^{p,q} \xrightarrow[\sigma_{p,q}]{\sim} E_\infty^{p,q} \xrightarrow[\theta^{p,q}]{\sim} \mathrm{Gr}_F^p H_C^{p+q} \cong \mathrm{Gr}_L^p H_X^{p+q} = \mathrm{Gr}_q^W H_X^{p+q}.$$

The first isomorphism uses Hodge theory; the second uses first-quadrant convergence. These are graded quotients, not a canonical direct-sum decomposition of the mixed Hodge structure H_X^n .

5. NATURALITY AND THE EQUIVARIANT COMPARISON

Define $T : C^\bullet \rightarrow C^\bullet$ by $T|_{K^{p,q}} := f_p^*$. The simplicial identities for $f_\bullet$ give $T\delta_1 = \delta_1 T$, and naturality gives $T\delta_2 = \delta_2 T$. Hence $TD = DT$ and $T(F^p C^\bullet) \subseteq F^p C^\bullet$. Write

$$\Phi := H^n(T) = f_\bullet^* : H_X^n \rightarrow H_X^n, \quad [a]_D \mapsto [Ta]_D.$$

Proposition 5.1 (Actions on the pages). *The filtered cochain map T induces, before using convergence, maps*

$$T_r^{p,q} : E_r^{p,q} \rightarrow E_r^{p,q}, \quad [a]_r \mapsto [Ta]_r.$$

Then these commute with d_r and the next-page isomorphism Ψ_{r+1} . In particular,

$$T_1^{p,q} = f_p^* : H^q(X_p, \mathbb{Q}) \rightarrow H^q(X_p, \mathbb{Q}),$$

$$T_2^{p,q}(\alpha + \mathrm{im} d_1^{p-1,q}) = f_p^* \alpha + \mathrm{im} d_1^{p-1,q} \quad (\alpha \in \ker d_1^{p,q}).$$

Proof. Preservation of F and commutation with D give $T(Z_r^{p,q}) \subseteq Z_r^{p,q}$ and $T(B_r^{p,q}) \subseteq B_r^{p,q}$. Thus T_r is well-defined, and

$$d_r^{p,q} T_r^{p,q} [a]_r = [DTa]_r = [TDa]_r = T_r^{p+q, q-r+1} d_r^{p,q} [a]_r.$$

Both orders of applying T and (1) send $[a]_{r+1}$ to $[Ta]_r + \mathrm{im} d_r^{p-r, q+r-1}$. Hence the action induced on $H^{p,q}(E_r, d_r)$ is exactly $T_{r+1}^{p,q}$ under Ψ_{r+1} . Proposition 2.2 gives the first two page formulas. $\square$

Since Φ preserves L , it induces

$$\Phi_L^p : \mathrm{Gr}_L^p H_X^n \rightarrow \mathrm{Gr}_L^p H_X^n, \quad x + L^{p+1} H_X^n \mapsto \Phi(x) + L^{p+1} H_X^n.$$

It preserves W and F_{Hdg} by naturality of the form comparison and Theorem 3.2. Under (2), $\Phi_L^p = \mathrm{Gr}_q^W(\Phi)$ for $n = p + q$.

Theorem 5.2 (The weight-graded action). *Let $X_\bullet, f_\bullet$ be as in Section 1, and let $n = p + q$. Then there is a canonical isomorphism of pure weight- q Hodge structures*

$$\alpha_{p,q} : \mathrm{Gr}_q^W H_X^n = \mathrm{Gr}_L^p H_X^n \xrightarrow{\sim} E_2^{p,q} = \frac{\ker d_1^{p,q}}{\mathrm{im} d_1^{p-1,q}},$$

and it satisfies $\alpha_{p,q} \mathrm{Gr}_q^W(\Phi) = T_2^{p,q} \alpha_{p,q}$.

Proof. Let $\rho_{p,q} : \mathrm{Gr}_L^p H_X^n \rightarrow E_\infty^{p,q}$ be the inverse of $\theta^{p,q}$ after transporting from F to L . We construct

$$\alpha_{p,q} := \sigma_{p,q}^{-1} \rho_{p,q}.$$

Its factors are isomorphisms by Theorem 2.3 and Proposition 4.2; Theorem 3.2 supplies their Hodge compatibility. For $a \in F^p C^n$ with $Da = 0$, put $\bar{a} = [a]_D + L^{p+1} H_X^n$. The representative rule for convergence gives

$$\rho_{p,q} \Phi_L^p(\bar{a}) = [Ta]_\infty = T_\infty^{p,q} \rho_{p,q}(\bar{a}).$$

By Proposition 5.1, the next-page identifications commute with T_r . Composing them gives $\sigma_{p,q} T_2^{p,q} = T_\infty^{p,q} \sigma_{p,q}$. Therefore

$$\alpha_{p,q} \Phi_L^p = \sigma_{p,q}^{-1} T_\infty^{p,q} \rho_{p,q} = T_2^{p,q} \alpha_{p,q}. \quad \square$$

For the explicit element rule, write a closed representative as

$$a = a_p + a_{p+1} + \cdots + a_n, \quad a_i \in K^{i, n-i}.$$

The first two components of $Da = 0$ give $\delta_2 a_p = 0$ and $\delta_1 a_p + \delta_2 a_{p+1} = 0$, with $a_{p+1} = 0$ if $p = n$. Thus $[a_p]_{\text{sing}} \in \ker d_1^{p,q}$, and

$$\alpha_{p,q}([a]_D + L^{p+1} H_X^n) = [a_p]_{\text{sing}} + \text{im } d_1^{p-1,q}.$$

The next-page identifications track this leading component to the stable class $[a]_\infty$. Independence of representatives follows because this formula is the constructed map $\sigma_{p,q}^{-1} \rho_{p,q}$. Applying T changes the leading component to $f_p^* a_p$, displaying the equivariance directly.

REFERENCES

- [SS] Companion notes, *From a double complex to convergence* ([spectral.tex](#)), and the [spectral-sequence course](#). We use their column filtration, cohomological indices, representative quotients, and induced-filtration convergence convention.
- [McC] J. McCleary, *A User's Guide to Spectral Sequences*, 2nd ed., Cambridge University Press, 2001, Definition 2.4 and Theorems 2.6, 2.15. [The two filtrations and the proof of Theorem 2.15](#).
- [D74] P. Deligne, *Théorie de Hodge, III*, Publ. Math. Inst. Hautes Études Sci. **44** (1974), 5–77. [\(8.1.19\)–\(8.1.20\)](#), pp. 34–35; [Proposition 8.3.5](#), p. 42.